

Label-efficient Land Cover Classification at High Spatial Resolution using Multi-stage Pre-training

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Figure 1: Tishomingo State Park

- From Tishomingo, MS
- Lived in Starkville since 2018
- Hobbies & interests: cycling, aviation, coding

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Academic background

- Graduated from MSU in 2021 with a B.S. in Computer Science (minor in mathematics)
- Started grad school in MSU Department of Sustainable Bioproducts in 2022
 - Research focus — deep learning for wood species identification
- Started PhD in Engineering with concentration in Biosystems Engineering in 2023
 - Research focus — deep learning for land cover classification at high spatial resolution
- January — June 2024: Machine Learning Scientist Co-op with Bayer Crop Science R&D
 - Analytic Pipeline Development — data science and AI for supporting decision-making in an advanced plant breeding pipeline
 - Specific investigations — improving spatial and temporal availability of key soil quality indicators using high spatial resolution satellite imagery and machine learning

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Land use and land cover (LULC)

- **Land cover:** biophysical material covering the Earth's surface (e.g., water, vegetation, soil, structures, etc.) [1] **“What is there?”**
- **Land use:** human interaction with the land cover (e.g., agriculture, forest practices, urban development, etc.) [2, 3] **“How is it being used?”**
- LULC data are used in a wide range of disciplines, including agriculture, forestry, urban planning, and environmental science [4–6]

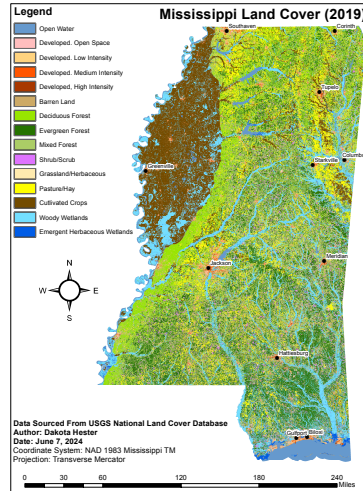


Figure 2: Land cover map of Mississippi.

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Classifying LULC

- Prior to the advent of GIS, LULC data was human-interpreted from aerial photography [2, 7].
- With advent of satellite remote sensing in 1980s, researchers and practitioners began using automated algorithms to classify LULC [8].
- Most classical approaches to LULC classification paradigms were pixel-wise statistical models that used coarse resolution multispectral data [9, 10].

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Pixel-wise classification

- Traditional classifiers usually operate by transforming a feature vector into output class probabilities for a given pixel.
- Common data sources include:
 - Surface reflectance data
 - Derived spectral indices (i.e., NDVI, EVI, NDWI, etc.)
 - Ancillary information (i.e., topography, soil, etc) [6, 11]
- Non-parametric, supervised classifiers are often used (e.g., Decision Trees, Random Forests, Support Vector Machines, etc.) [9, 10].

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Pixel-wise classification

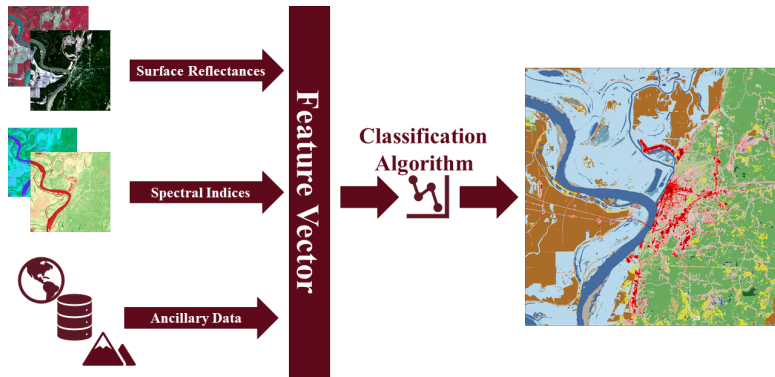


Figure 3: Pixel-wise classification workflow

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Land cover at very high spatial resolution

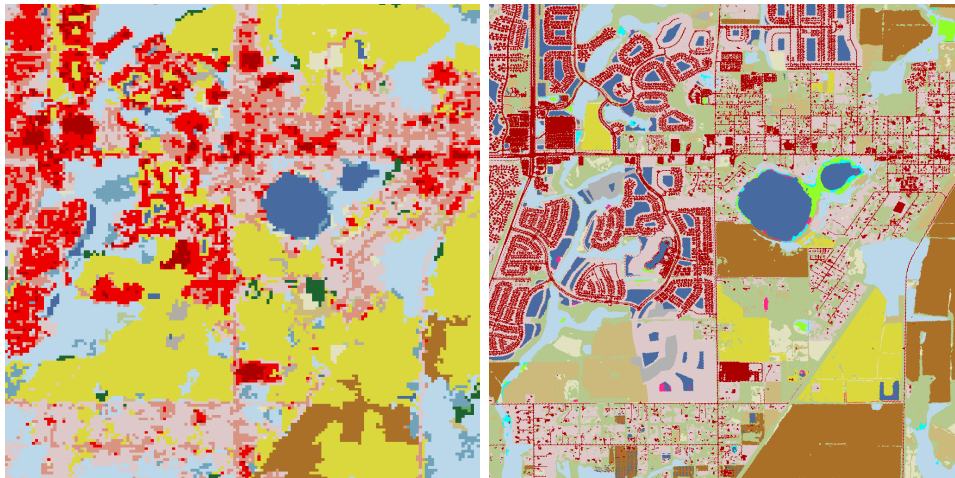


Figure 4: Comparison of land cover maps at 30 m (left) and 1 m (right) spatial resolution

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Decreased Class Discriminability

Classifying pixels gets **more difficult** as spatial resolution **increases** [12, 13].

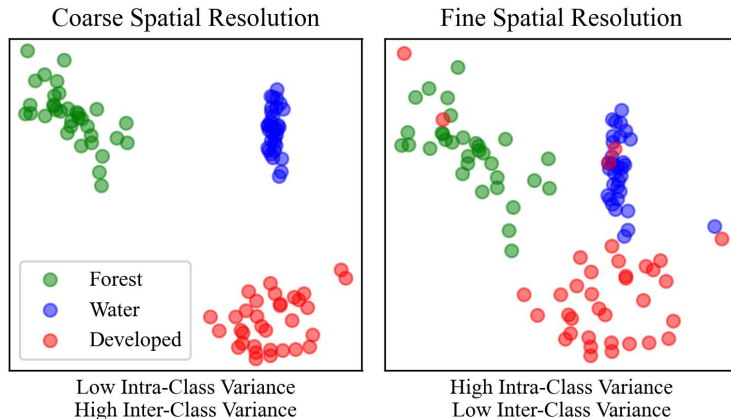


Figure 5: As spatial resolution increases, inter-class variance decreases [13].

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Decreased Class Discriminability

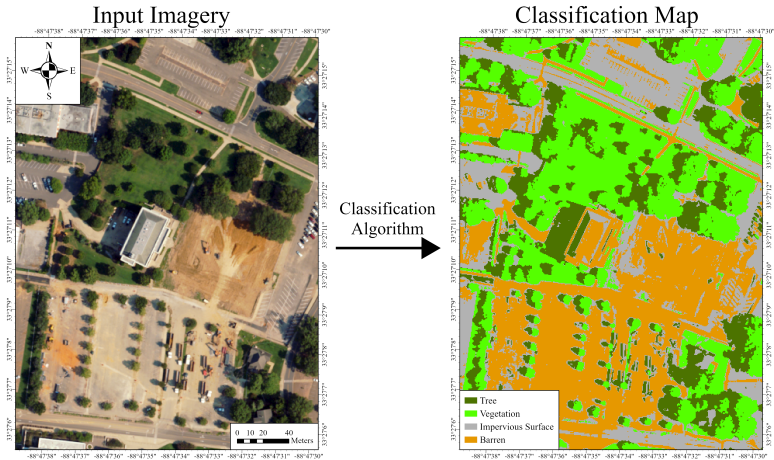


Figure 6: Applying a pixel-wise classifier to high spatial resolution data typically results in noisy classification maps.

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Object-Based Image Analysis (OBIA)

- One approach to classifying high spatial resolution data is to use **segmentation** to group neighboring pixels that share similar spectral characteristics into **objects**.
- Instead of classifying individual pixels, the classifier operates on the objects.
- Object-based image analysis (OBIA) addresses local variance concerns and produces cleaner, more accurate classification maps [14, 15].

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Object-Based Image Analysis (OBIA)

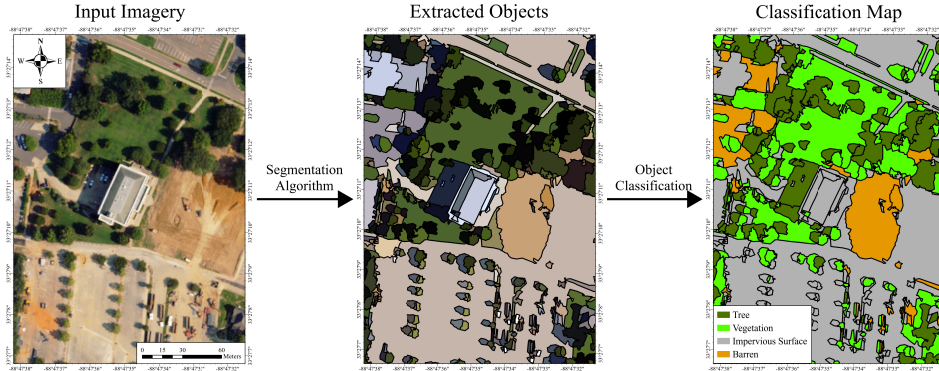


Figure 7: OBIA workflow.

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- OBIA is more complex, and requires more human supervision that pixel-wise classification methods [16].
- Ideally, we would like to have a generalizable method that can produce accurate classification maps without the need for extensive human intervention.

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“Why do we need deep models?”

- Deep learning models are more complex, resource intensive, and difficult to train compared to traditional classifiers [17].
- However, deep models have several advantages:
 - ❶ **Feature extraction** — deep models can learn complex **spatial** hierarchies of features that are difficult to engineer by hand [18].
 - ❷ **Flexibility** — the nature of deep learning architectures allows them to be applied to a wide range of tasks across different modalities, and can be modified to best suit a given task [19].
 - ❸ **End-to-end learning** — unlike traditional methods, deep models can be trained in an end-to-end fashion, where little to no pre-processing or post-processing is required [16, 20–22].

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Spatial Feature Extraction through Convolution

- Convolutional filters emphasize local spatial patterns images.
- A small filter moves across input imagery, performing a dot product with the input regions it covers at each location.
- The result is a **feature map** that highlights the presence of certain spatial patterns at different locations in the input image.
- The behavior of the filter is determined by its size, stride, and its parameters.

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Convolutional Filter

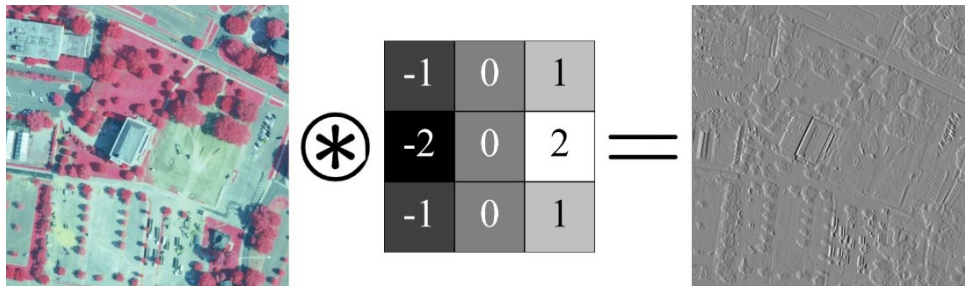


Figure 8: A convolutional operation on an aerial image. This convolutional filter highlights vertical lines.

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- Multiple convolutional filters can be applied to an input to extract feature maps that highlight different spatial patterns.
- These feature maps can then be passed to more convolutional filters to extract more complex patterns.
- The parameters of the convolutional filters are learned through an iterative process, where the gradient of a loss function that measures the difference between the predicted output and the true output with respect to the parameters is used to update the parameters [23, 24].

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Convolutional Neural Networks

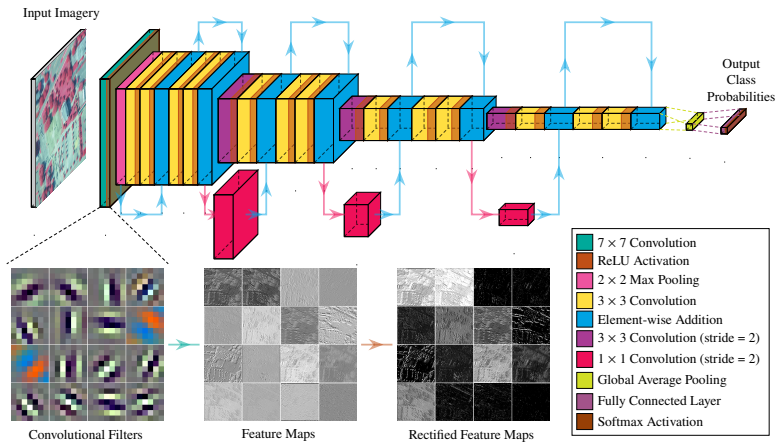


Figure 9: A ResNet-18 CNN architecture [25]. A sampling of learned convolutional filters from the first layer is shown.

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- An naive approach to land cover classification would be to assign a single label to each tile that is passed through a CNN, where this label represents the dominant land cover class in the tile [26–28].
- However, the feature maps learned by the CNN inherently **spatially structured**, and these feature maps can be leveraged to produce dense, pixel-wise classification maps through a process called **semantic segmentation**.

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- In an encoder-decoder architecture, an **encoder** CNN extracts features from the input such that a **decoder** network can upsample the features to the original input resolution [29].
- The decoder network learns to generate a pixel-wise classification map from the features extracted by the encoder.
- The most popular encoder-decoder network Architecture is U-Net[30].

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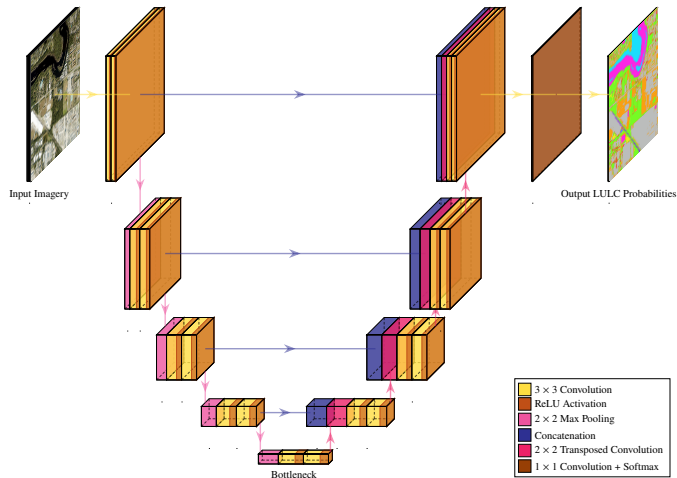


Figure 10: U-Net architecture [30].

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- While deep learning networks are powerful, their primary drawback is the amount of data required to train them.
- For example, ResNet-50, a popular CNN architecture, has over 25,000,000 parameters that need to be learned [25].
- For land cover, this is especially problematic, as high resolution data is complex and difficult to annotate at scale [31].
- One approach to mitigate this issue is to use **transfer learning**, where a network can be **pre-trained** on a related task with a large amount of data, then **fine-tuned** on the target task with a smaller amount of data [32].

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Transfer Learning

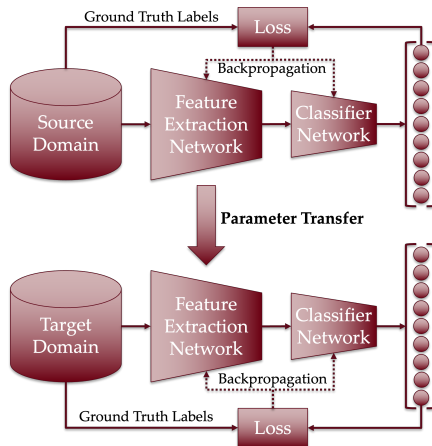


Figure 11: Transfer learning paradigm.

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Denoising Autoencoder

- While labeled data is optimal for pre-training, some pre-training tasks can be performed in a **self-supervised** manner, where the network learns to predict some property of the input data without requiring labels [33].
- Denoising autoencoders are a type of encoder-decoder architecture that are trained to remove noise from an input image [34].
- In effect, noise is added to an input image, then train the network to predict the original image from the noisy image.
- The network needs to learn **features** that are **robust** to noise while being meaningful for reconstruction [35].

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Our goal

- Our overarching goal is to develop a 1m resolution land cover map that covers the entire state of Mississippi with a high degree of accuracy.
- Data annotation is expensive and time-consuming, so the main focus of our work is devising a creative pre-training strategy that combines self-supervised pretraining with a supervised fine-tuning stage to reduce the amount of labeled data required for training a deep model.
- We construct a large U-Net model that undergoes various pre-training regimens, then fine-tune the model on only **119** tiles of fully annotated data from the state of Mississippi.

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- Our U-Net architecture is based on the original U-Net architecture proposed by Ronneberger, et al. [30].
- In place of the original encoder network, we use a ResNet-152 architecture [25].
- Overall, our complete U-Net architecture has over 100,000,000 parameters, making it a powerful model, but also difficult to train — especially in the absence of large amounts of data.

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- We want to evaluate how effective a large-scale unsupervised pre-training stage can be in reducing the amount of labeled data required for training a deep semantic segmentation model.
- During the first pre-training stage, 3 scenarios will be evaluated: a model pre-trained on Imagenet, a model pre-trained using a denoising autoencoder, and a randomly initialized model.
- The implications of an additional stage of pre-training will also be evaluated. Under this scenario, a much smaller dataset of fully labelled data will be used to further pre-train the model.
- Finally, the model will be fine-tuned on a very small dataset of fully labelled data from the state of Mississippi.
- The performance of the model will be evaluated on a separate test set of fully labelled data from the state of Mississippi.

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USDA National Agricultural Imagery Program

- USDA NAIP imagery covers CONUS with 3-year revisit cycle at 0.6 m spatial resolution.
- NAIP imagery is available in 4 spectral bands (RGB + NIR).
- Our study uses 2023 NAIP imagery over Mississippi clipped
 - Color infrared composites (Red, Green, and NIR bands) resampled to 1m spatial resolution
 - Imagery is clipped into tiles of 256x256 pixels
 - 20% of images places in pretraining set, and 5% in pretraining validation set (377,913 and 94,480 tiles, respectively)
 - For fine-tuning, 250 tiles total were selected to be labelled (119 in training set, 61 in validation set, and 70 in test set)

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Chesapeake Bay Land Cover Data

- In order to enhance the model's ability to fit the data, we introduce a second pre-training stage using fully annotated data from the Chesapeake Bay Land Cover dataset.
- 2018 NAIP tiles and corresponding land cover data extracted from Fairfax County, VA.
- Why Fairfax County?
 - High density of hard-to-classify impervious land cover types (i.e., roads, buildings, etc.)
- Why not directly apply these tiles to the fine-tuning stage?
 - The land cover legend is for Chesapeake Bay — for example, it lacks a Cultivated Crops class.
 - High resolution ancillary canopy height data was used to construct the data, which we do not have for Mississippi.
 - Generalization of the model to Mississippi is not guaranteed with data from Chesapeake Bay Watershed.
- 10,220 tiles used for training and 5,110 used for validation.

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Our Denoising Autoencoder

- Our encoder network of choice is a ResNet-152 architecture.
- For the Denoising Autoencoder, we extract feature maps produced by each of the four stages of the encoder network, resample these to the original input size, concatenate them, and pass them through a single convolutional 1×1 convolutional layer to produce the input reconstruction.
- While the shallow decoder network inhibits the ability of the network to produce high quality reconstructions, it helps ensure that the encoder learns to extract features.
- To speed up training, our encoder network has been pre-trained on ImageNet, a large corpus of “natural” images.
- To evaluate the efficacy of this stage, we compare the performance of our self-supervised model to one that has been only been pre-trained using ImageNet as well as a randomly initialized model.

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Fully Supervised Pre-training Stage

- Once the encoder has been trained, we place it in a U-Net architecture and train the network on the Fairfax County land cover data.
- One important note is that we have the option to **freeze** the encoder during this stage. In doing so, we can reduce the amount of data required to train the full network, but lose the some ability to learn a more task-specific feature representation.
- In our case, we found that there was enough data in the Fairfax County dataset to train the full network without freezing the encoder parameters.

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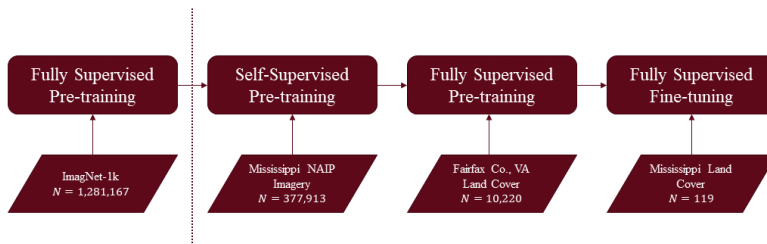


Figure 12: Model training flowchart

Pre-training with a denoising autoencoder improved the performance of the model under the 2-stage pre-training strategy.

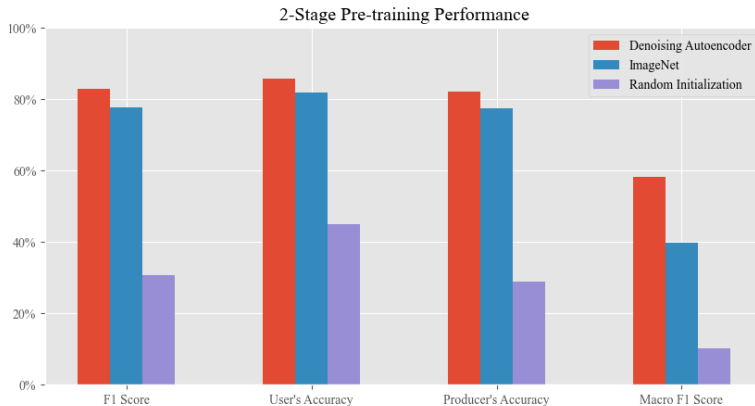


Figure 13: Performance of models on the Mississippi land cover dataset.

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Pre-training with a denoising autoencoder improved the performance of the model under the 2-stage pre-training strategy.

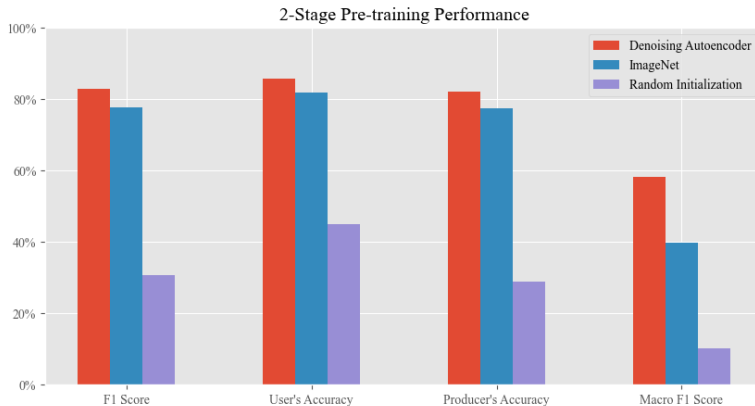


Figure 14: Performance of models on the Mississippi land cover dataset.

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Confusion Matrix

True Label	Open Water	81.2%	3.9%	0.5%	0.5%	0.2%	1.6%		12.1%
	Impervious (raised structures)	0.0%	67.3%	6.6%	4.3%	0.4%	15.8%		5.5%
	Impervious (paved surfaces)	0.0%	1.7%	80.0%	5.8%	1.1%	9.3%	0.0%	2.1%
	Barren Land	0.1%	0.7%	7.5%	45.6%	1.0%	35.2%	8.5%	1.4%
	Tree Canopy/Woody Vegetation	0.0%	0.0%	0.1%	0.0%	91.2%	3.7%	1.0%	3.9%
	Low Vegetation/Herbaceous	0.0%	0.2%	2.1%	5.0%	5.0%	73.3%	11.8%	2.6%
	Cultivated Crops			0.1%	1.5%	3.2%	23.5%	71.5%	0.3%
	Unclassified	0.1%	0.5%	2.0%	0.2%	8.1%	4.7%	0.1%	84.3%
		Open Water	Impervious (raised structures)	Impervious (paved surfaces)	Barren Land	Tree Canopy/Woody Vegetation	Low Vegetation/Herbaceous	Cultivated Crops	Unclassified

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Warren County Land Cover

GCER Lab | Warren County Mississippi 2023 Land Cover ⓘ

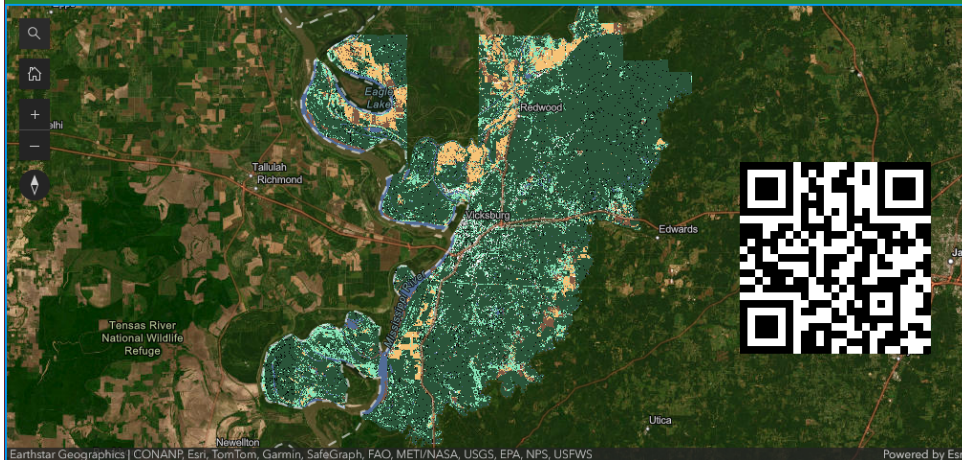


Figure 16: Warren County Land Cover Map.

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- Large scale pre-training of a deep model using an unsupervised regimen can enable the training of an over-paramaterized deep model on a small amount of labeled data.
- Further, additional pre-training using a modest amount of labeled data can boost the performance of the model even further.
- Still, classifying more visually intricate land cover classes remains a challenge, even with a large model.

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- More data — clear need to improve the accuracy of cultivated crops and barren land classes.
- Post-processing using OBIA-inspired techniques to refine classification map edges.
- Applying model to super-resolved satellite imagery.

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